Blueprint — ELI Eval Harness

Contents

Exact structural guide to `tools/eval/`. The harness drives ELI's **real pipeline** and asserts on its **own telemetry** (action, matched_by, grounding_confidence, response_mode, latency) plus answer text. 100% local, offline by default, no cloud judge, no required extra deps (PyYAML only, already a project dep).

> Two tools share one driver: **`run_eval.py`** (behavioural PASS/FAIL board) and **`profile_runtime.py`** (runtime-graph profiler — replays the stored conversations + cases under a `sys.settrace` line tracer to report exact line-% hit, action coverage, which of the 15 agents fire, and a hot/cold module heatmap with a ranked cold-file cut-list). Full run commands for both are in `tools/eval/README.md`.

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1. Directory tree

```
tools/
├── __init__.py                    # makes `tools` importable
└── eval/
    ├── __init__.py
    ├── eli_driver.py              # THE shared core — only file that calls ELI
    ├── assertions.py              # deterministic checks over a driver result
    ├── cases.yaml                 # the test cases (data, not code)
    ├── run_eval.py                # pure-Python green/red board (canonical)
    ├── profile_runtime.py         # runtime-graph profiler (what executes / dead code)
    ├── README.md                  # quickstart
    └── promptfoo/                 # optional Node UI front-end
        ├── eli_provider.py        # thin adapter → eli_driver
        └── promptfooconfig.yaml
```

Two front-ends (`run_eval.py`, `promptfoo/`) sit over **one** driver (`eli_driver.py`). Neither front-end knows how to call ELI; only the driver does. Add a third front-end by importing the driver — never by re-implementing the call.

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2. The contract everything depends on: the normalised result dict

Both driver entry points return a plain `dict` with this shape. Assertions and the promptfoo provider read **only** these keys, so the rest of the harness is insulated from ELI's internal return shapes.

key	type	router	engine	meaning
`target`	str	-	-	`"router"` or `"engine"`
`action`	str	-	-	routed/executed action (e.g. `NEXT_MEDIA`)
`matched_by`	str	-	-	router rule id (e.g. `media.app_targeted`)
`confidence`	float	-	-	route / response confidence
`args`	dict	-	—	router action args
`text`	str	`""`	-	the answer shown to the user
`grounding`	float | None	None	-	`trace.grounding_conf`
`response_mode`	str	`""`	-	`meta.response_mode` (e.g. `ungrounded_hedge`)
`latency_s`	float	-	-	wall-clock for the call
`skipped`	bool (engine)	—	-	True when no model → case SKIPs, not fails
`raw`	original	-	-	the untouched router dict / process() result

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3. File-by-file contracts

`eli_driver.py` — the shared core

The only module that imports ELI internals.

- `route_only(prompt: str, network: Optional[bool]=None) -> dict` Calls `eli.execution.router_enhan` **No model required.** Fast (ms). Returns the normalised dict with `target="router"`.
- `run_engine(prompt, network=None, reasoning_mode="quick", session_id=None) -> dict` Runs the full `CognitiveEngine.process(...)`. **Needs a model loaded.** Normalises str / dict / generator returns. On engine-init failure returns `{"skipped": True, "reason": "no engine/model"}` so cases SKIP, not crash.
- `get_engine()` — lazily builds & caches one headless `CognitiveEngine( auto_init_gguf=True)`; returns None if it can't init. Model comes from `config/settings.json` (swap there to compare models).
- `_network(state)` — context manager that monkeypatches `eli.core.config.network_allowed` for the call **only**, then restores it. **Non-persisting** — running the eval never flips the user's real offline setting. `state=None` leaves it as-is.

`assertions.py` — deterministic checks

- `check(assertion: dict, result: dict) -> (passed: bool, detail: str)` No LLM judge. Reads only the normalised result dict.

type	value	passes when
`contains` / `not_contains`	str	(case-insensitive) substring of `text`
`regex`	pattern	`re.search` on `text`
`action_is` / `action_not`	str | list	`action` ∈ / ∉ list
`matched_by`	str	substring of `matched_by`
`response_mode`	str | list	`response_mode` ∈ list
`grounding_min` / `grounding_max`	float	`grounding` ≥ / ≤ value
`hedged`	bool	answer is an honest “won’t guess” hedge
`max_latency_s`	float	`latency_s` ≤ value

`cases.yaml` — the test data

List of case dicts:

```yaml
- id: next_track_real          # unique
  target: router               # router (fast, model-free) | engine (needs model)
  prompt: "next track"
  network: on                  # optional: on|off — forced for this case only
  mode: quick                  # optional engine reasoning mode (default quick)
  assert:
    - {type: action_is, value: [NEXT_MEDIA, MEDIA_CONTROL]}
```

`run_eval.py` — the board

CLI over the driver. Flags: - `--target router|engine|all` (default `router`) - `--cases <path>` (default `cases.yaml` next to the script) - `--filter <substr>` (match on case `id`) - `--json <path>` (machine-readable records)

Prints PASS/FAIL/SKIP per case + failing-assertion detail + a summary line. **Exit code is non-zero if any case fails** → drop into CI / pre-push.

`promptfoo/eli_provider.py` — optional Node front-end

`call_api(prompt, options, context) -> {"output": text, "metadata": {action, matched_by, grounding, response_mode, latency_s, skipped}}`. Thin adapter that calls `eli_driver.run_engine`. Network/mode via test `vars`.

`promptfoo/promptfooconfig.yaml`

Declares the provider + sample tests. Deterministic text asserts work out of the box; routing/grounding asserts read `metadata`. **Keep any `llm-rubric` judge pointed at a LOCAL model — never a cloud API.**

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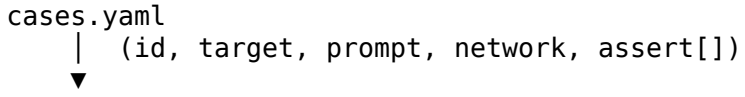
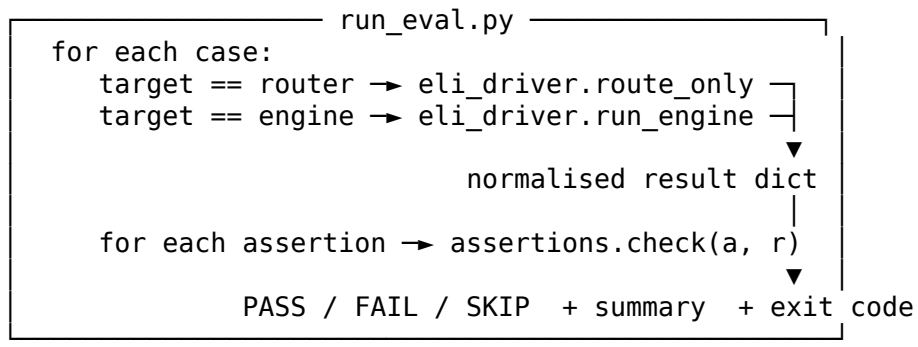
4. Data flow





`promptfoo eval → eli_provider.call_api → eli_driver.run_engine → same dict`

`route_only` and `run_engine` are the only doors into ELI. Everything upstream (cases, assertions, boards, promptfoo) is plumbing over the normalised dict.

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5. How to extend

Add a case — append to `cases.yaml`. Rule of thumb: *every bug a log reveals becomes a case.* Pick `target: router` if it's a routing decision (fast, no model); `target: engine` if it needs an answer / grounding / latency.

Add an assertion type — add a branch in `assertions.py::check` returning `(bool, detail)`. It automatically reads the normalised dict, so no driver change is needed.

Surface a new ELI signal — add the key in `eli_driver` (both `route_only` and/or `run_engine` return dicts), then assert on it. Example: to assert on `agents_used`, read it from `raw` in the driver and expose it as a top-level key.

Compare two models — set model A in `config/settings.json`, run `--target all --json a.json`; swap to model B, `--json b.json`; diff. Watch `grounding`, `hedged`, `correctness`, `latency_s`. This is how "is the 24B worth the CPU latency on this box" becomes a number.

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6. Run commands

```
cd ~/Desktop/ELI_MKXI-main_MAY_NEWEST && source .venv/bin/activate

python tools/eval/run_eval.py                       # router board, ~60ms, no model
python tools/eval/run_eval.py --target all          # + engine cases (needs a model)
python tools/eval/run_eval.py --filter media        # subset by id
python tools/eval/run_eval.py --target all --json out.json
python tools/eval/run_eval.py || echo "EVAL FAILED"   # pre-push guard

# optional Node UI:
cd tools/eval/promptfoo && npx promptfoo@latest eval && npx promptfoo@latest view
```

Sanity: the bare `run_eval.py` should print `10 passed  0 failed  0 skipped`. Import error ⇒ not at repo root or venv not active.

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7. Design invariants (don't break these)

1. **One driver.** Only `eli_driver.py` imports ELI internals. Front-ends go through it.
2. **Normalised dict is the contract.** Assertions/providers never touch ELI's raw return shapes except via `raw`.

3. **Network state is per-call and non-persisting.** Never write `network_enabled` to settings from the harness.
4. **No cloud judge.** Deterministic assertions first; any LLM-graded check uses a local model. Offline by default.
5. **Engine cases degrade to SKIP**, never crash, when no model is present — so the router board always runs in CI.
6. **Cases are data.** New coverage = new YAML, not new code (unless it's a new assertion type or signal). "'